Nonlocality in image denoising and other variational problems

Department of Mathematics, University of Pavia

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Book of Abstracts

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1 Invited Talks

A prox-Based Semi-Smooth Newton Method for Convex Variational Problems

Speaker: Soeren Bartels (University of Freiburg)

Abstract: In this talk, we present a prox-based semi-smooth Newton method for a broad class of convex variational problems with non-differentiable energy densities, including TV-minimisation, p-Dirichlet type energies, elasto-plastic torsion, and obstacle-type problems. The idea is to reformulate the subdifferential optimality conditions using proximity operators, which yields a nonlinear operator equation with a Newton-differentiable structure. Under mild local strong convexity assumptions on the energy density, we establish local superlinear convergence based on invertibility and uniform boundedness of Newton derivatives. The framework recovers classical semi-smooth Newton schemes for obstacle-type problems and exhibits a primal-dual invariance. Numerical examples illustrate the efficiency of the method for representative model problems.

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Some variational problems in non-local frameworks: embeddings, free-discontinuity problems, and homogenization

Speaker: Giuseppe Cosma Brusca (SISSA, Trieste)

Abstract: We present some results concerning classical problems in the Calculus of Variations formulated in various non-local frameworks. We begin by introducing non-local Sobolev spaces by means of a Gagliardo-like seminorm, discussing their continuous embeddings together with possible applications to geometric variational problems. Subsequently, we describe the non-local-to-local passage by illustrating several Γ-convergence results. First, we consider non-local energies in the context of phase transitions and free-discontinuity problems. Then, we address the multiscale analysis of energies that oscillate periodically or that are subject to Dirichlet boundary conditions on perforated domains, showcasing different behaviours in accordance with the specific non-local setting. Based on joint works with Konstantinos Bessas, Andrea Braides, Francesco Cassandra, Davide Donati, Giuliana Fusco, Sergio Scalabrino, Chiara Trifone, and Edoardo Voglino.

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Continuity of critical points of Riesz energies in dimension 1

Speaker: Davide Carazzato (University of Vienna)

Abstract: We will present a regularity result for critical points of Riesz energies in dimension 1 for a general class of interaction kernels assuming that the critical points admit bounded density with respect to the Lebesgue measure. Under some convexity assumptions on the kernel, we provide a control on the oscillations of the density, yielding

to the continuity result. This result was obtained in collaboration with N. Fusco and A. Pratelli.

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Some properties of the solutions of Total-Variation regularized inverse problems

Speaker: Antonin Chambolle (CEREMADE Dauphine/PSL University)

Abstract: Together with M. Łasica (Warsaw), we revisited the issue of jump set inclusion for solutions of total-variation denoising problems (aka ROF problem). We proposed a quite general approach which allowed to show the result in many different settings (convex data terms, vectorial/color images, nuclear norm total variation...), including, in a recent extension with K. Bessas (Pavia), non-local, such as fractional total variations.

Joint works with Michał Łasica (Warsaw) and Konstantinos Bessas (Pavia).

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Periodic minimizers for nonlocal interaction energies

Speaker: Lucia De Luca (CNR, Rome)

Abstract: We consider systems of particles on the real line interacting through negative Gagliardo seminorms. We prove that the 1-periodic configuration is the only minimizer of the energy and that the gradient flow converges exponentially fast to such a ground state. We consider also the case of Riesz type interactions. Joint works with M. Goldman, G. Pini, M. Ponsiglione, F. Santilli, E.N. Spadaro.

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A fast multipole method for fractional total variation

Speaker: Patrick Dondl (University of Freiburg)

Abstract: We consider variational problems of the form

$$\min_u \ \frac{1}{2}\int_\Omega\int_\Omega \frac{|u(x)-u(y)|}{|x-y|^{d+s}}\,\mathrm{d}x\,\mathrm{d}y \ + \ \Lambda\int_\Omega |u-g|^p\,\mathrm{d}x,$$

where the regularizer is the s-fractional total variation, i.e. the seminorm of $W^{s,1}(\Omega)$. In contrast to models built on a squared fractional Sobolev seminorm, the associated Euler-Lagrange equation is a nonlinear nonlocal equation involving the fractional 1-Laplacian, so Fourier spectral methods are unavailable. Standard fast summation fails for a more basic reason: the first variation

$$\sum_j G(|x_i - x_j|)\ \mathrm{sign}(u_i - u_j)$$

carries a charge that depends on the target index, which destroys the source-target decoupling fast multipole methods rely on.

We present a discretization that Γ-converges to the continuous model, together with an algorithm that restores the FMM structure. The key observation is that sorting the degrees of freedom by value converts the sign nonlinearity into an insertion order: each gradient entry becomes a kernel-weighted sum over the already-inserted set, so the kernel seen by the multipole expansion is linear again.

This is joint work with Konstantinos Bessas (Pavia) and Jan-Philip Kniesel (Freiburg).

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Generalized BMO-type seminorms and vector-valued Sobolev functions

Speaker: Serena Guarino Lo Bianco (University of Modena and Reggio Emilia)

Abstract: We establish a pointwise limit theorem for a broad class of parameter-dependent BMO-type seminorms as the parameter tends to zero. By introducing novel BMO-type seminorms, we extend several existing results on characterization of Sobolev-type spaces to the vector-valued setting. Precisely, for any open set $\Omega \subset \mathbb{R}^n$ and any $p \in (1, \infty)$, we establish a non-distributional characterization of both the Sobolev space $W^{1,p}(\Omega; \mathbb{R}^m)$ and the space $E^{1,p}(\Omega; \mathbb{R}^n)$ of L^p maps with p-integrable distributional symmetric gradient.

Moreover, for all $p \in [1, \infty)$, we show that these seminorms converge to integral functionals with convex, p-homogeneous integrands associated with the distributional gradient and the symmetric gradient.

The talk is based on a joint work with Konstantinos Bessas (Università di Pavia) and Roberta Schiattarella (Università degli Studi di Napoli Federico II).

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Some observations on the Gamow liquid drop model

Speaker: Dario Mazzoleni (University of Pavia)

Abstract: In this talk, we will deal with the (generalized) Gamow liquid drop model ($N \geq 2$):

$$\min \left\{ P(\Omega) + \int_\Omega \int_\Omega \frac{1}{|x-y|^{N-\alpha}} \, \mathrm{d}x\mathrm{d}y : \Omega \subset \mathbb{R}^N,\ |\Omega| = m \right\},$$

focusing mostly on the nonexistence issue for large masses m. We will show some elementary geometric strategies to obtain some interesting information on this problem.

This is a joint ongoing work with Aldo Pratelli (Pisa) and Berardo Ruffini (Bologna).

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On the Minimisation of Nonlocal Interaction Energies

Speaker: Maria Giovanna Mora (University of Pavia)

Abstract: Nonlocal interaction energies play a central role in describing the collective behaviour of large particle systems in a wide range of applications. In this lecture we will focus on interaction energies that are short-range repulsive and long-range attractive. We will review the key results on the existence and uniqueness of minimisers, and present their explicit characterisation in the classical case of isotropic kernels with Riesz-type repulsion. We will then show how a complete characterisation can be given for a broad class of anisotropic variants of the repulsive kernel. This talk is based on joint works with several collaborators: R. Frank, J. Mateu, L. Rondi, A. Scagliotti, L. Scardia, E.G. Tolotti, and J. Verdera.

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To be announced

Speaker: Matteo Novaga (University of Pisa)

Abstract: *To be announced*

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An improved quantitative fractional isoperimetric inequality

Speaker: Berardo Ruffini (University of Bologna)

Abstract: We present an improved version of the quantitative fractional isoperimetric inequality, in which a stronger notion of asymmetry appears. In particular, we show that the square root of the isoperimetric deficit controls, not only the Fraenkel asymmetry, but also a suitable notion of oscillation of the boundary. This extends the result in the local setting obtained by Fusco and Julin in 2011. The proof follows a similar spirit but is different with respect to that in the local case. This is a joint work with E. Cinti and E. M. Merlino.

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Isoperimetric sets and profile monotonicity in planar convex bodies

Speaker: Giorgio Saracco (University of Ferrara)

Abstract: Given a planar convex set K, we provide a novel proof of the characterization of isoperimetric sets in K. It relies on a suitable Legendre-type duality with the prescribed-curvature functional, which also yields regularity properties of the isoperimetric profile. Relying on this novel approach, we prove that the isoperimetric profile is monotone along

the volume-preserving parallel-body flow. This is based on joint work with Fattah, Ftouhi, and Leonardi.

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BMO-Type Functionals for Image Denoising: Anisotropic Total Variation and Γ-Convergence

Speaker: Roberta Schiattarella (University of Naples Federico II)

Abstract: We study an anisotropic family of BMO-type functionals based on local mean oscillations over disjoint translations of a fixed reference set. Their pointwise limit on SBV functions distinguishes the absolutely continuous and jump parts of the derivative through two different anisotropic densities. Since the pointwise limit may fail to exist for general BV functions, we consider their Γ-convergence in L^1 and with respect to stronger topologies. We also discuss applications to image denoising, where these functionals act as scale-dependent regularizers and their Γ-limits describe the effective variational models. The talk is based on joint works with Fernando Farroni, Nicola Fusco, Serena Guarino Lo Bianco, and Luigi Greco.

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Some perspectives on the BBM formula

Speaker: Giorgio Stefani (University of Padua)

Abstract: We present some recent results on the Bourgain-Brezis-Mironescu (BBM) formula, including sharp conditions for its validity, semigroup analogues, and weighted versions. These results were obtained in collaboration with L. Gennaioli, and with A. Kubin and G. Saracco.

References

[1] Luca Gennaioli and Giorgio Stefani. "Sharp conditions for the BBM formula and asymptotics of heat content-type energies". In: *Arch. Ration. Mech. Anal.* 250.1 (2026), Paper No. 8, 46. ISSN: 0003-9527,1432-0673. DOI: 10.1007/s00205-025-02157-1. Preprint available at: https://arxiv.org/abs/2502.14655.

[2] Andrea Kubin, Giorgio Saracco, and Giorgio Stefani. "On the Γ-limit of weighted fractional energies". In: *Proceedings of the Royal Society of Edinburgh: Section A Mathematics* (2025), pp. 1–22. DOI: 10.1017/prm.2025.10100. Preprint available at: https://arxiv.org/abs/2503.19875.

[3] Giorgio Stefani. "On a weighted version of the BBM formula". In: *J. Math. Anal. Appl.* 557.2 (2026), Paper No. 130272, 18. ISSN: 0022-247X,1096-0813. DOI: 10.1016/j.jmaa.2025.130272. Preprint available at: https://arxiv.org/abs/2504.06736.

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2 Contributed Talks

A Nonlocal Total Variation prior for image processing: Variational and PDE perspectives

Speaker: Francesc Alcover Borràs (Universitat de les Illes Balears)

Abstract: In this talk, we present a weighted nonlocal total variation prior for image processing, extending the nonlocal total variation introduced by Kindermann et al. (2005). We first introduce the associated seminorm and the corresponding space of functions of nonlocal bounded variation from a functional-analytic prespective, establishing several structural results and properties relevant to optimization. We then study the prior within a nonlocal variant of the Chambolle–Lions model. Under suitable assumptions on the weight function and the fidelity operator, existence and uniqueness of minimizers are established. To compute solutions numerically, we develop primal–dual and TV-flow algorithms. Finally, we investigate the behavior of the proposed prior through a series of experiments. In particular, we analyze the asymptotic behavior of the nonlocal TV flow, characterize piecewise-affine functions as those which are least penalized by the prior, and compare its performance with other related image processing priors on representative image restoration tasks.

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Homogenization effects on non-local functionals

Speaker: Enrico Micalizio (Politecnico di Torino)

Abstract: Non-local functionals are a powerful tool to model long-range interactions. In this talk, we study the macroscopic behavior of a class of non-local functionals featuring a rapidly oscillating periodic weight. By means of homogenization and Γ-convergence, we will reveal how the interplay between periodicity and non-locality forces the minimizing sequences to develop highly oscillating microstructures. As a natural consequence, we establish that the effective macroscopic functional fails to admit a standard double-integral representation.

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A general compactness result for localization of nonlocal gradients

Speaker: Felix Seifert (KU Eichstätt-Ingolstadt)

Abstract: Localization results provide a rigorous way to relate nonlocal models to their local counterparts. In the context of variational models, Γ-convergence together with the necessary compactness results offers the natural framework for analyzing such limit processes. In this work, we discuss this approach for problems in which nonlocality arises through nonlocal gradients, as in models of nonlocal hyperelasticity. Whereas previous

works often focused on kernels resulting from a specific construction, such as rescaling, or on special classes, like fractional kernels, we instead consider arbitrary sequences of radial kernels that satisfy suitable localization, monotonicity, and regularity conditions. This yields a more general approach that both encompasses various earlier results and extends them further, in particular by allowing for kernels with unbounded support and even non-integrable kernels. The main technical difficulty lies in the proof of compactness, which relies on establishing a nonlocal Poincaré inequality with uniform constants, as well as a uniform continuous embedding into a nonlocal Sobolev space given by a fractional reference kernel. To illustrate the scope of our results, we will show how different relevant examples fit into this setting. This is joint work with Carolin Kreisbeck (KU Eichstätt-Ingolstadt) and Carlos Mora-Corral (Universidad Auónoma de Madrid).

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Homogenization in one-dimensional higher-order non-local models of phase transitions

Speaker: Edoardo Voglino (SISSA, Trieste)

Abstract: We study the limit behavior of Cahn-Hilliard-type functionals in which the derivative is replaced by higher-order fractional derivatives and modulated by an oscillating factor. Depending on the ratio between the oscillation scale and the interface length, we identify three different regimes and prove Γ-convergence in each regime to a suitable sharp-interface limit functional. In the extreme regimes, we prove a separation-of-scales effect that enables us to highlight the difference relative to the local models.

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